Conserved Thermal Strain, Accurate Reactions, and Remaining Displacement Error

Structural Verification Lab

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Abstract

How do thermal contact resistance and transfer between nonmatching meshes affect a restrained two-material bar? An original, reproducible software study connects conservative finite-volume heat conduction, exact overlap integration of thermal strain, and C++17 linear finite-element assembly. Independent reference problems separate implementation agreement, time error, smooth-field spatial error, and reconstruction error. Four-level studies recover first-order backward Euler and second-order smooth-field spatial convergence. A 45-case steady contact/property sweep gives nearly exact reactions while retaining local displacement error. Thus conservation of the integral controlling reaction force does not certify the local temperature reconstruction. These are synthetic numerical verification results, not physical validation or a new discretization claim.

1 Question, scope and governing model

The study extends the existing Structural Verification Lab; its six imported component snapshots and original beam study remain available with pinned provenance [3]. The new research question concerns one-way thermal-to-mechanical coupling, motivated by heat/interface-resistance topics in DTU Modelling in Materials and Manufacturing Engineering and eligible TUM finite-element topics. No private course files, exercise data or teaching code are distributed.

Consider a bar on $[0, L]$ with constant area A, piecewise constant conductivity k, volumetric heat capacity c, Young's modulus E, and expansion coefficient α. Let $\theta = T - T_{\mathrm{ref}}$ be the increment from a stress-free temperature. With no distributed axial force, the governing equations and an independent compatibility identity are

$$cA\dot{T} - \partial_x(kA\partial_x T) = As, \qquad N' = 0, \tag{1}$$

$$N = EA(u' - \alpha\theta), \qquad N = \frac{u_R - u_L - \int_0^L \alpha\theta\, dx}{\int_0^L (EA)^{-1}\, dx}. \tag{2}$$

Positive N denotes tension, and support forces on the bar are $(-N, N)$. Displacement follows by integrating $N/(EA)+\alpha\theta$ from the left support. Thermal contact has areal resistance $R \geq 0$ in m^2 K/W and jump $T^- - T^+ = q''R$ for positive rightward heat flux q''. Mechanical bonding remains perfect: temperature may jump but displacement does not. Both restrained and traction-free mechanical endpoints are supported.

The assumptions are small strain, quasi-static linear elasticity, positive constant material properties within each cell/element, and prescribed heat sources or boundary temperatures. There is no mechanical inertia, thermal feedback from deformation, latent heat, plasticity, contact opening, or temperature-dependent constitutive response. Conductivities and capacities in the smooth transient experiment are deliberately synthetic; they are not identified material data.

2 Connected discretization and conservation contract

Cell-centred heat unknowns represent cell averages. For neighbouring cells of lengths h_i, h_{i+1}, the conductance and backward-Euler update are

$$g_{i+1/2} = \frac{A}{h_i/(2k_i) + R_{i+1/2} + h_{i+1}/(2k_{i+1})}, \qquad (M + \Delta t H)T^{n+1} = MT^n + \Delta t b^{n+1}, \tag{3}$$

where $M_{ii} = c_i A h_i$, H is the resistance-network matrix, and b includes sources and boundary contributions. Shared face rates cancel in the global energy balance. This construction uses standard finite-volume conservation principles [1].

For thermal cells I_i and mechanical elements J_e, exact interval overlaps define the transferred element mean,

$$\bar{\theta}_e = |J_e|^{-1} \sum_i |I_i \cap J_e| \theta_i, \qquad \sum_e \alpha_e |J_e| \bar{\theta}_e = \int_0^L \alpha(x) \theta_h(x)\, dx. \tag{4}$$

This is a one-dimensional, piecewise-constant instance of conservative mesh transfer; general mesh projection is established work [2]. Mechanical material jumps must be mechanical nodes and thermal faces. Both meshes share identical endpoints and constant area. A purely thermal contact face may lie inside a mechanical element. These explicit alignment conditions make the material-weighted identity valid without smoothing a temperature jump.

For a constant-property P1 element of length ℓ_e,

$$K_e = \frac{E_e A}{\ell_e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \qquad f_e^{\mathrm{th}} = E_e A \alpha_e \bar{\theta}_e \begin{bmatrix} -1 \\ 1 \end{bmatrix}. \tag{5}$$

The original C++17 kernel assembles stiffness and thermal load through a C ABI. Python owns arrays, applies boundary elimination including nonzero prescribed displacements, and solves with NumPy. Heat operators, overlap integration and reference solvers also remain in Python. There is no speedup claim. Since N in Eq. (2) depends on one global eigenstrain integral, exact transfer can recover its value for a supplied cell field while leaving local reconstruction error in $u(x)$.

3 Independent references and separated refinement studies

All values below come from versioned experiment JSON, with tables generated automatically. The bar has $L = 1$ m, $A = 2 \times 10^{-4}$ m^2, and a material interface at $x = 0.4$ m. Unless varied explicitly, $(E_1, E_2) = (70, 190)$ GPa and $(\alpha_1, \alpha_2) = (23, 11) \times 10^{-6}$ K^{-1}. Both mechanical ends are fixed. The mechanical mesh has three left and five right elements.

For smooth refinement, take insulated heat boundaries, $k = c = 1$ in SI units, zero contact resistance, and initial increment $40\cos(\pi x)$ K. Heat initial data are exact cell averages; the continuum solution is $40e^{-\pi^2 t}\cos(\pi x)$ at $t = 0.1$ s. Distinct expansion coefficients ensure that its zero unweighted mean still produces a nonzero reaction. Analytic sine antiderivatives provide continuum force and displacement independently of transfer and assembly; Gaussian quadrature separately checks these integrals.

The temporal experiment fixes 20 heat cells and compares backward Euler with exact semidiscrete evolution obtained by diagonalizing $M^{-1/2} H M^{-1/2}$. The reference shares the spatial heat operator, so this comparison isolates time error, not heat spatial error. An unequal-mass two-cell closed form and the uniform-grid cosine eigenvalue independently check the spectral solver. The spatial experiment uses this exact semidiscrete evolution on 10–80 cells and compares with continuum cell averages and analytic mechanics. Thus time error cannot contaminate its measured spatial order.

Study	Steps/cells	e_T (K)	e_N (N)	e_u (m)	p_T
Time	10	6.931e-01	5.700e+01	3.218e-06	–
Time	20	3.535e-01	2.907e+01	1.641e-06	0.9714
Time	40	1.786e-01	1.468e+01	8.289e-07	0.9854
Time	80	8.974e-02	7.379e+00	4.166e-07	0.9926
Space	10	1.191e-01	9.918e+00	5.599e-07	–
Space	20	3.014e-02	2.478e+00	1.399e-07	1.9829
Space	40	7.556e-03	6.194e-01	3.497e-08	1.9958
Space	80	1.890e-03	1.548e-01	8.742e-09	1.9989

Table 1. Maximum temperature error e_T, maximum element force error e_N, and maximum error at mechanical nodes e_u. Time rows count steps; space rows count cells. $p_T = \log_2(e_{T,\mathrm{coarse}}/e_{T,\mathrm{fine}})$.

Temporal force and nodal orders also approach one (0.9714, 0.9854, 0.9926). Spatial force and nodal orders are approximately 2.0008, 2.0002 and 2.0001. These orders apply to this smooth, zero-contact problem. The fixed mechanical mesh retains between-node P1 interpolation error; nodal convergence does not measure its removal. A separate affine-temperature mechanics study recovers the exact midpoint interpolation error $\alpha\beta h^2/8$ for $\theta = \beta x$.

4 Transfer, thermal contact and the global–local distinction

Frozen-field transfer. Fix 7+11 thermal cells, with $\theta = 80 - 50x$ K at left cell centres and $\theta = 10 + 20x$ K at right cell centres. Reconstruct these values as constants on their cells and refine only the mechanical target grid. The independent source-cell integral is 0.0008024 m and its reaction is −18088 N. The comparison baseline linearly interpolates cell-centre values at element midpoints, bridging the jump and clamping beyond the end centres; it represents a different field.

Elements	Integral error	Overlap force error	Naive force error
5	1.351e-16	2.011e-16	2.011e-16
9	2.702e-16	6.034e-16	4.023e-16
17	1.351e-16	4.023e-16	3.931e-03
33	4.054e-16	3.017e-15	5.267e-03

Table 2. Material-weighted integral error is normalized by the sum of absolute source-cell strain integrals (floor 10^{-12} m). Force errors are relative to the source-field reference.

Overlap preserves the controlling integral to roundoff. The naive baseline is also accurate on the two coarse grids because of cancellation, then reaches about 0.53% force error. This nonmonotone result is retained: no universal superiority or convergence rate is inferred from these four points.

Transient contact demonstration. Use the same 18 heat cells and eight mechanical elements, $(k_1, k_2) = (13, 4)$ W/(m K), $(c_1, c_2) = (2, 3)\times 10^5$ J/(m^3 K), $T_{\mathrm{ref}} = 293.15$ K, and an initially uniform reference temperature. Fix end temperatures to 373.15 and 293.15 K; run twenty 500 s steps for each $R \in \{0, 0.02, 2\}$ m^2 K/W. At 10000 s the computed compressive forces are respectively −19136.55, −18692.40 and −16690.30 N. These are discrete model outputs, not converged transient contact predictions. Across 60 steps, the maximum source-field force error is 1.7462e-15 on scale $\max(1\,\mathrm{N}, |N_{\mathrm{ref}}|)$; the energy defect is 2.1074e-12 on scale $\max(1\,\mathrm{J}, |\Delta U|, |Q_{\mathrm{boundary}}\Delta t|)$. The mechanics reference integrates original thermal cells, independently of transferred means.

Steady contact/property sweep. To separate contact physics from time error, impose boundary increments 80 and 0 K and compute steady solutions on the same grids. Vary k_2/k_1 and E_2/E_1 independently over $\{0.1, 1, 10\}$ with $k_1 = 13$ W/(m K), $E_1 = 70$ GPa, and five resistances. The 45 cases use the independent series-resistance solution

$$q'' = \frac{80}{0.4/k_1 + R + 0.6/k_2}, \qquad Q = Aq'', \qquad [T] = q''R. \tag{6}$$

Integrating each material's affine temperature in Eq. (2) provides an independent continuum mechanics reference.

R (m^2 K/W)	Heat-rate rel. error	Force rel. error	e_u (m)
0.0e+00	5.103e-14	1.190e-14	1.451e-06
1.0e-10	8.583e-14	9.807e-15	1.451e-06
2.0e-02	6.454e-14	1.052e-14	9.272e-07
2.0e+00	4.217e-13	1.180e-14	9.697e-08
1.0e+05	2.170e-08	1.172e-14	2.417e-12

Table 3. Worst errors across nine property pairs at each resistance. The displacement column is an absolute nodal error, not an equilibrium residual.

Near-exact global reactions coexist with nodal error up to 1.4512e-06 m. On these uniform-within-material heat grids, full material integrals of an affine field are exact, but partial integrals ending inside a thermal cell are not. High-resistance boundary rates lose relative accuracy through subtraction of nearly equal temperatures; the maximum heat-matrix condition number is 1.8521e+03. This bounded sweep does not establish the infinite-resistance limit.

5 Reproducibility, evidence and limitations

The original extension lives in `thermomechanics/` and `native/`. Python dataclasses enforce SI material and mesh contracts; the C ABI checks native buffer dimensions. The existing six component snapshots remain byte-checked against `suite.json`. Their original histories and licenses are preserved. The original integration code and new extension are MIT licensed. The earlier beam synthesis remains available at the v0.1.0 tag [3]; component papers are supporting studies, not substitutes for this synthesis.

From the repository root, create a Python environment and run the following commands. The README supplies compiler setup and a short native-build-path option for Windows. A working C++17 toolchain is required; a missing native library fails explicitly.

```
python suite.py install
python suite.py verify
python suite.py native
python suite.py test
python -m thermomechanics.coupled_experiment
python -m thermomechanics.refinement_experiment
python manuscript.py --write-tables
python suite.py paper --engine tectonic
python manuscript.py
```

The numerical baseline is revision `c87b8203bec3` [3]. The coupled and refinement JSON files record complete inputs, all histories/cases and normalized-LF SHA-256 hashes of Python/native sources. The paper build checks those source hashes and refuses stale generated tables before compiling. Its manifest binds the manuscript, generator, build runner, numerical evidence, all numerical sources, snapshot manifest and resulting PDF. It rejects unresolved references, overfull boxes and more than five pages. CI independently compiles with pdfLaTeX and uploads the PDF, log and manifest. Hash equality establishes provenance; it does not replace independent numerical checks or visual page review.

The preceding numerical revision passed 368 Python checks and five native CTests locally and on Linux and Windows. Both hosted numerical artifacts were downloaded and checked against their reported digests. These counts describe that numerical revision; the manuscript change adds separate stale-evidence mutation checks. The full suite checks free expansion, uniform restraint, nonzero prescribed endpoints, Python/C++ assembly agreement, energy balance, independent force identities and experiment reproduction. Cross-platform tolerances account for small floating-point residuals. For the steady sweep, relative force and scaled temperature gates are 10^{-11}; heat-rate tolerance is 10^{-10} except 2×10^{-7} at $R = 10^5$ m^2 K/W. The smooth-order gates are 0.95–1.02 in time and 1.9–2.1 in space.

Four kinds of evidence must remain distinct. First, Python/C++ and transfer checks test implementation consistency. Second, time error is measured against the same semidiscrete operator. Third, smooth spatial and steady contact cases use independent continuum references. Fourth, physical validity would require independently justified constitutive data and experiments; none are claimed. In particular, the heterogeneous transient contact demonstration has no continuum refinement result here. The steady and smooth transient studies cover complementary parts of the model and cannot be combined into that missing claim.

The implementation uses dense heat and mechanics solves and a dense overlap matrix. It targets small verification problems, with no large-scale performance, nonlinear response or arbitrary-interface convergence guarantee. Exact reaction force is insufficient as a sole acceptance metric: local displacement, supplied-field consistency, energy balance and independent continuum errors reveal different failures. A future extension could test transient continuum contact accuracy with independent interface eigenmodes, but that remains outside this release's evidence.

6 Conclusion

The connected Python/C++ pipeline makes the thermal-strain conservation contract explicit and testable. Its measured results support first-order time accuracy, second-order smooth spatial accuracy, and bounded steady-contact verification under the stated assumptions. Most significantly for verification practice, accurate global reactions survive a reconstruction that still produces local displacement error. The reproducible evidence therefore treats global conservation as one necessary check within a broader error analysis.

References

[1] R. Eymard, T. Gallouët and R. Herbin. Finite volume methods. *Handbook of Numerical Analysis*, vol. 7, 2000. Author reference: https://perso.math.u-pem.fr/reymard/.

[2] P. E. Farrell and J. R. Maddison. Conservative interpolation between volume meshes by local Galerkin projection. *Computer Methods in Applied Mechanics and Engineering* 200(1–4), 89–100, 2011. doi:10.1016/j.cma.2010.07.015.

[3] Structural Verification Lab. Original numerical baseline, revision `c87b8203bec3`, 31 October 2026; earlier beam synthesis at tag v0.1.0. https://github.com/racostacme-come/structural-verification-lab. Exact source/data hashes accompany this manuscript.